

Credit Application Governance Analysis

Data Ecosystems and Governance in Organizations

MSc in Business Analytics, Nova SBE

Team 13

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Agenda

01

Data Quality Assessment

Identifying & remediating issues in the raw dataset

02

Bias and Fairness Analysis

Disparate Impact, age-based patterns & proxy discrimination

03

Privacy and Governance

Sensitive data mapping, pseudonymization & GDPR compliance

04

Governance Recommendations

Controls, audit trails & AI Act alignment

01. Data Quality Assessment

502

Raw Records

2

Duplicates Found

6

Issue Categories

499

Clean Records

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Raw Records

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Duplicates Found

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Clean Records

6

Issue Categories



Duplicate Records

app_001 and app_042 duplicated — flagged with DUPLICATE_ENTRY_ERROR and removed



Missing Values

email (7), DOB (4), IP (4), SSN (4), ZIP (1) across 500 records



Gender Code Abbreviations

114 records (22.7%) — M/F vs Male/Female + 2 empty values set to Unknown



Inconsistent Date Formats

161 records (32.3%) — mixed ISO, DD/MM/YYYY, YYYY/MM/DD



Income Data Type Mismatch

financials.annual_income: 3 types found (float, int, str) — standardised to float



Negative/Invalid Values

credit_history < 0: app_043 and app_156 corrected to 0.
app_290 savings -5000 removed

01. Data Quality Assessment



Duplicate Records

app_001 and app_042 duplicated — flagged with DUPLICATE_ENTRY_ERROR and removed

Uniqueness

```
--- VERSION 2 (Internal _id:
{'_id': ObjectId('69ac60b262bbbc9a5048aff5'), 'app_id':
'app_001', 'applicant_info': {'email':
'stephanie.nguyen47@mail.com', 'full_name': 'Stephanie
Nguyen'}, 'decision': {'loan_approved': False,
'rejection_reason': 'high_dti_ratio'}, 'financials':
{'annual_income': 102000, 'credit_history_months': 37,
'debt_to_income': 0.42, 'savings_balance': 0}, 'notes':
'DUPLICATE_ENTRY_ERROR', 'spending_behavior': [{ 'amount':
576, 'category': 'Fitness' } ] }
```

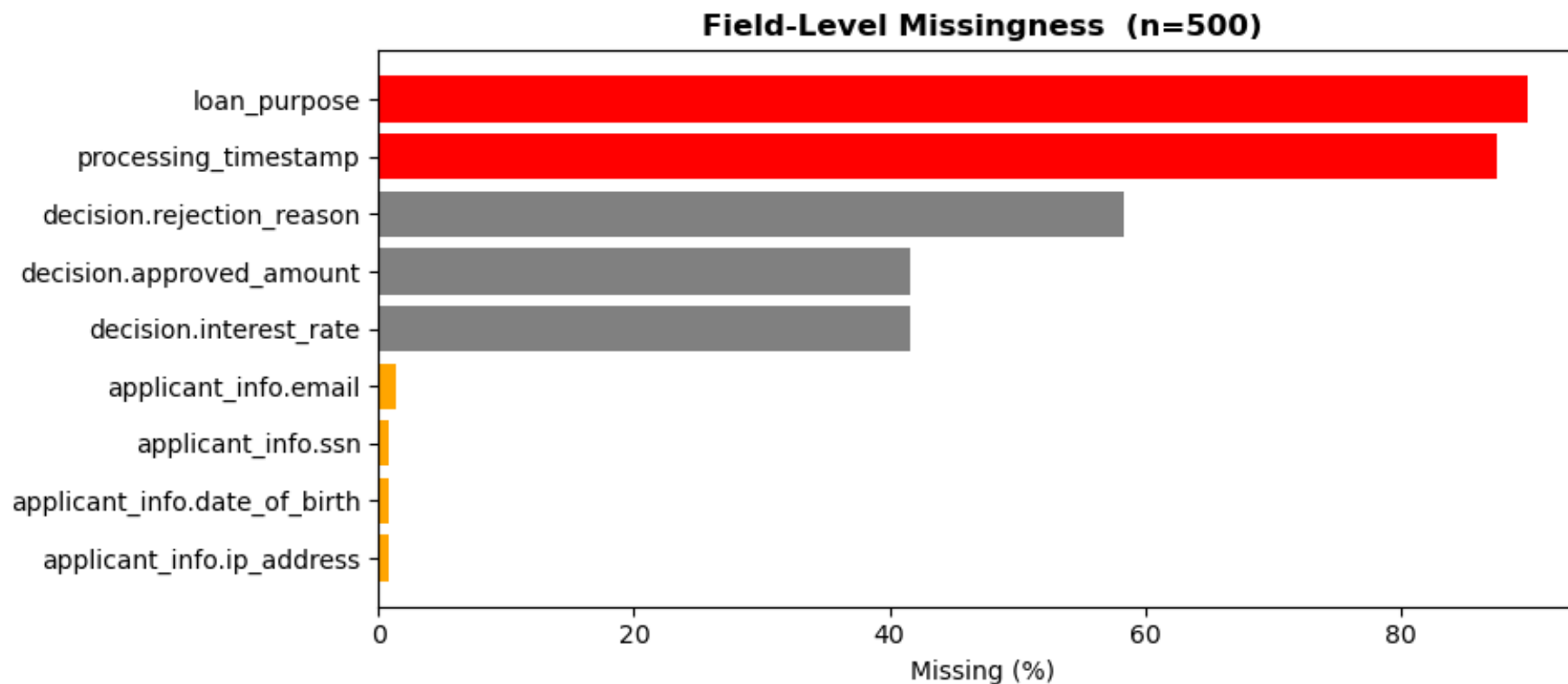
01. Data Quality Assessment



Missing Values

email (7), DOB (4), IP (4), SSN (4), ZIP (1) across 500 records

Completeness



01. Data Quality Assessment

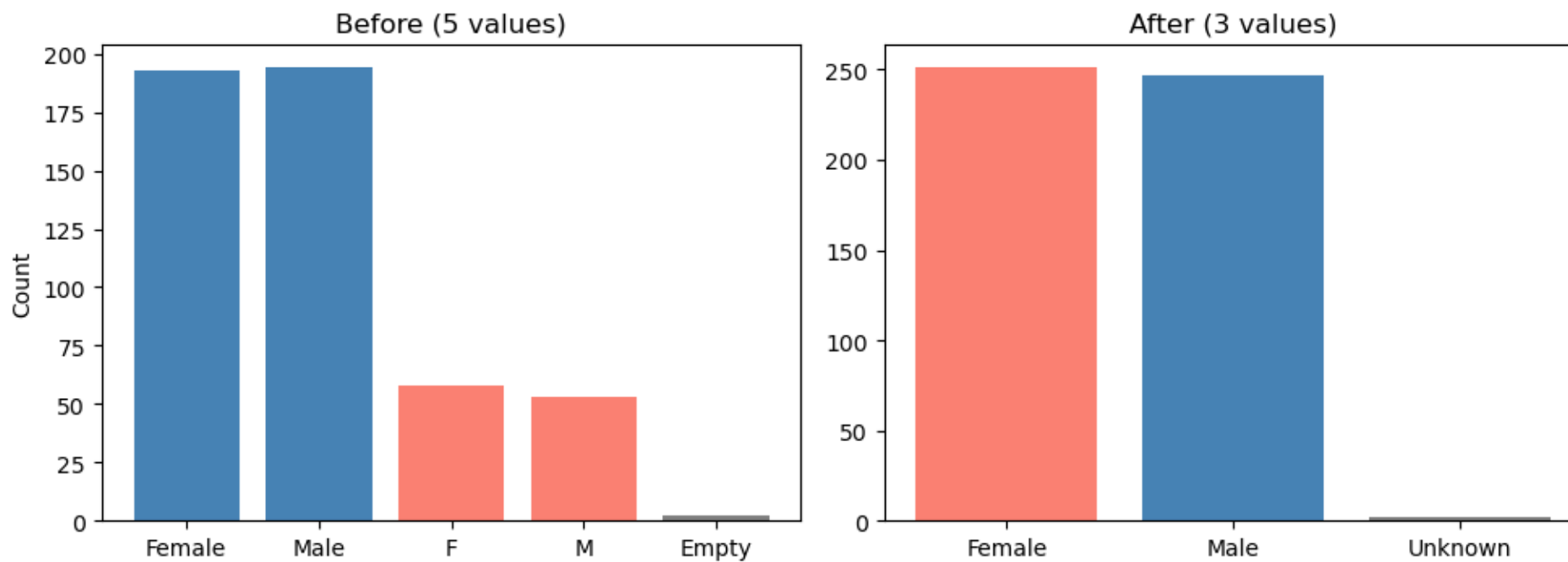


Gender Code Abbreviations

114 records (22.7%) — M/F vs Male/Female + 2 empty values set to Unknown

Consistency

Gender Encoding: Before vs After



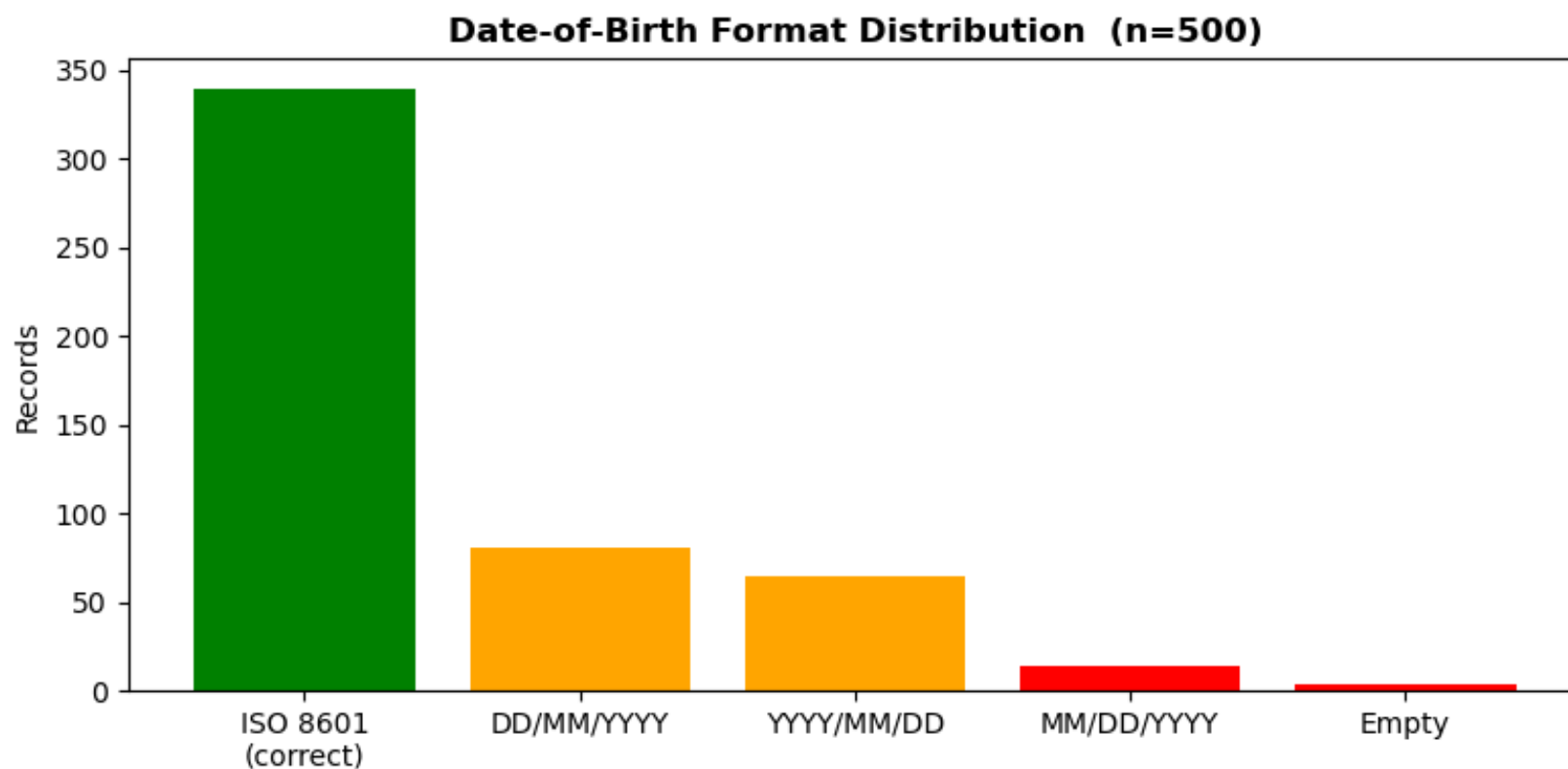
01. Data Quality Assessment



Inconsistent Date Formats

161 records (32.3%) — mixed ISO, DD/MM/YYYY, YYYY/MM/DD

Consistency



01. Data Quality Assessment



Income Data Type Mismatch

financials.annual_income: 3 types found
(float, int, str) — standardized to float

Validity



Negative/Invalid Values

credit_history < 0: app_043 and app_156 corrected to 0.
app_290 savings -5000 removed

```
DATA TYPES FOUND -----
applicant_info.date_of_birth | str
applicant_info.email | str
applicant_info.full_name | str
applicant_info.gender | str
applicant_info.ip_address | str
applicant_info.ssn | str
applicant_info.zip_code | str
decision.approved_amount | int
decision.interest_rate | float
decision.loan_approved | bool
decision.rejection_reason | str
financials.annual_income | float, int, str
financials.annual_salary | int
financials.credit_history_months | int
financials.debt_to_income | float
financials.savings_balance | int
loan_purpose | str notes | str processing_timestamp | str
```

```
No negative values in financials.annual_income.
Found 2 records with negative
financials.credit_history_months:
- App ID: app_043 | Value: -10
- App ID: app_156 | Value: -3
No negative values in financials.debt_to_income.
Found 1 records with negative financials.savings_balance:
- App ID: app_290 | Value: -5000
No negative values in decision.interest_rate.
No negative values in decision.approved_amount.
```

01. Data Quality Assessment



Income Data Type Mismatch

financials.annual_income: 3 types found
(float, int, str) — standardized to float

Validity



Negative/Invalid Values

credit_history < 0: app_043 and app_156 corrected to 0.
app_290 savings -5000 removed

```
DATA TYPES FOUND -----
applicant_info.date_of_birth | str
applicant_info.email | str
applicant_info.full_name | str
applicant_info.gender | str
applicant_info.ip_address | str
applicant_info.ssn | str
applicant_info.zip_code | str
decision.approved_amount | int
decision.interest_rate | float
decision.loan_approved | bool
decision.rejection_reason | str
financials.annual_income | float, int, str
financials.annual_salary | int
financials.credit_history_months | int
financials.debt_to_income | float
financials.savings_balance | int
loan_purpose | str notes | str processing_timestamp | str
```

No negative values in financials.annual_income.

Found 2 records with negative

financials.credit_history_months:

- App ID: app_043 | Value: -10

- App ID: app_156 | Value: -3

No negative values in financials.debt_to_income.

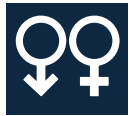
Found 1 records with negative **financials.savings_balance:**

- App ID: app_290 | Value: -5000

No negative values in decision.interest_rate.

No negative values in decision.approved_amount.

02. Bias and Fairness Analysis



Gender Disparate Impact

Female approval rate 50.4% vs. Male 66.0%
DI = 0.76, below 0.80 threshold → FAIL (AI Act Art. 10)



Spending Category Proxy

Gambling, Adult Entertainment and Alcohol flagged as sensitive model inputs: potential proxy discrimination (Art. 10 §5)



Age-Based Discrimination

18–24 cohort approved at 41.7% vs. 65.3% for 35–44
DI = 0.556 → worst performing age band



Opaque Rejection Reasons

algorithm_risk_score accounts for majority of rejections – no meaningful explanation provided to applicants



Geographic Redlining Risk

Approval rates differ across LA, NYC and Atlanta. ZIP code acts as proxy for race/ethnicity (AI Act Recital 44)



Intersectional Discrimination

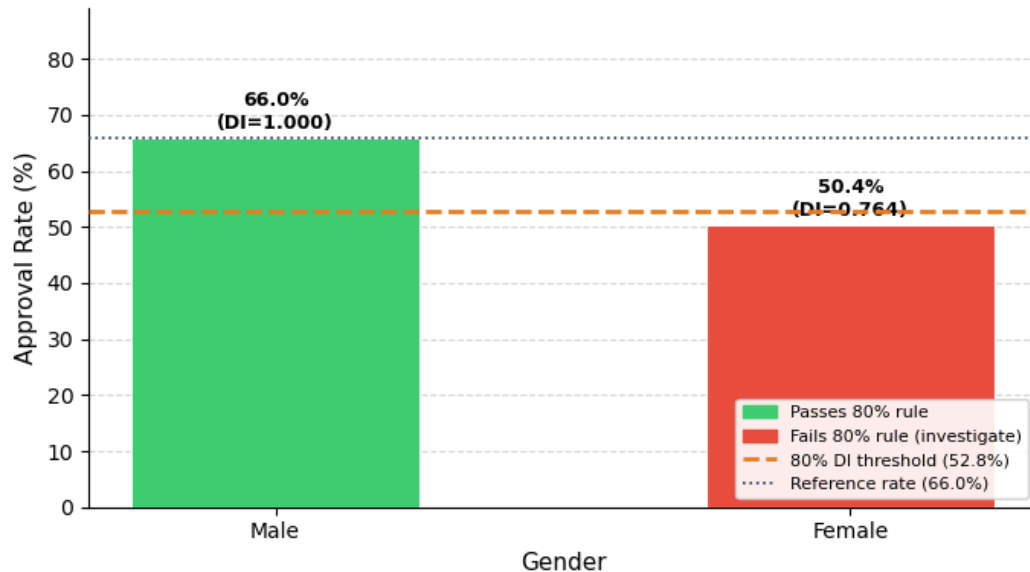
Female / 18–24 is the most at-risk subgroup – compounding gender and age bias confirmed (AI Act Recital 44)

02. Bias and Fairness Analysis

Gender Disparate Impact Ratio

Four-Fifths Rule ($DI < 0.8$ = potential discrimination)

**Loan Approval Rate by Gender
with EEOC Four-Fifths (80%) Disparate Impact Threshold**

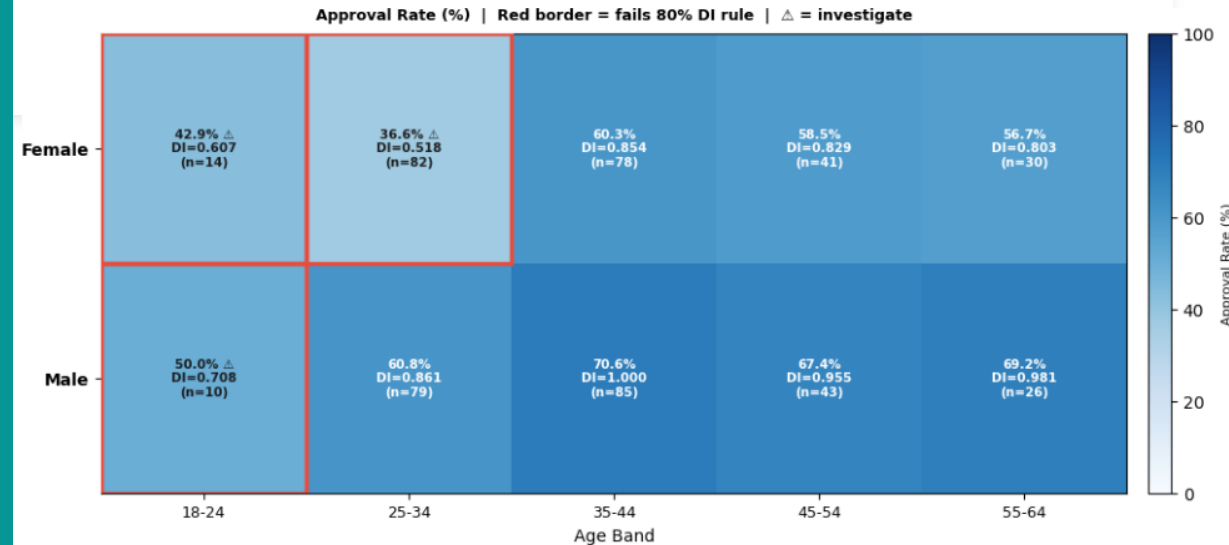


DI = 0.76

Below 0.80 threshold

Female 50.4% vs Male 66.0%
→ Both fail the 80% rule

Gender Disparate Impact Ratio



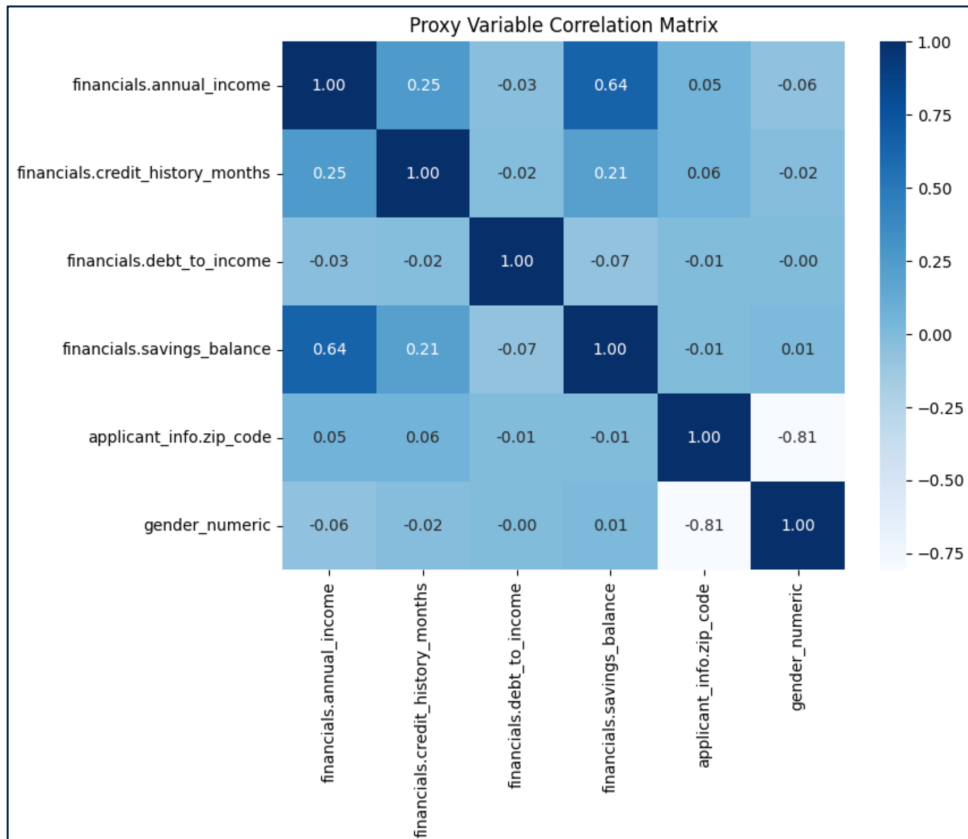
DI = 0.556

Below 0.80 threshold

Female applicants aged 18–25 approved at 41.7% vs. 65.3% for Male / 35–44

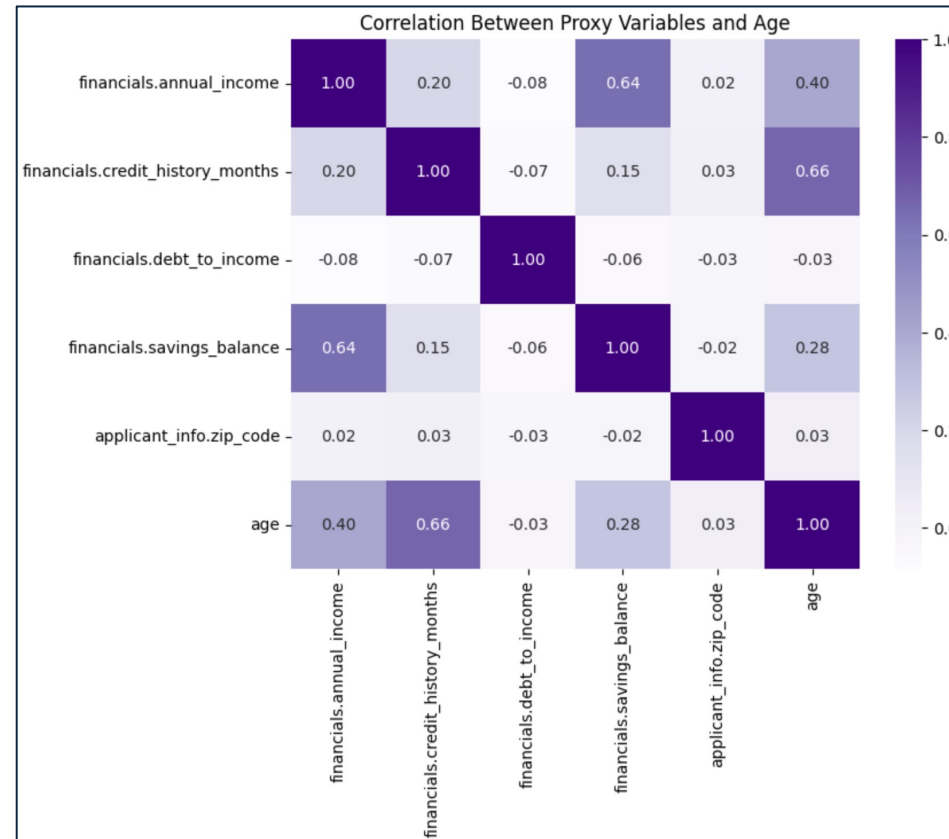
→ Intersectional discrimination confirmed (AI Act Recital 44)

03. Privacy and Governance – Proxy Analysis



Correlation of non-protected variables with gender

- ZIP code (-0.81) has the strongest link to gender.
- It is the main potential gender proxy in the dataset.
- All other tested variables show near-zero correlation with gender.
- ZIP code should be reviewed carefully before model use



Correlation of non-protected variables with age

- Credit history months (0.66) has the strongest link to age.
- It is the main potential age proxy in the dataset.
- Annual income (0.40) and savings balance (0.28) also show some age-related pattern.
- ZIP code (0.03) and debt-to-income (-0.03) show near-zero correlation.

Correlations with loan approval are overall weak, but some variables may still raise fairness concerns.
 * More detailed interpretation of these relationships is provided in the notebook.

03. Privacy and Governance



Personal Identifiable Information (PII):

Direct Identifiers

full_name	HIGH
email	HIGH
ssn	CRITICAL
ip_address	HIGH

Indirect Identifiers

gender	HIGH
date_of_birth	HIGH
zip_code	MED
spending_behavior	HIGH

Financial Personal Data

credit_history_months	MED
debt_to_income	HIGH
savings_balance	HIGH
annual_salary	HIGH

03. Privacy and Governance

Privacy Demonstration: Pseudonymization, Anonymization and Data Minimization

Step 1: Pseudonymization of direct identifiers

SHA-256 Hashing applied to:

- Full name
- Email
- SSN
- IP Address (and removal)

GDPR Art. 5(1)(c) - personal data should be adequate, relevant, and limited to what is necessary. Hashing reduces direct identifiability, but it is still pseudonymization, not full anonymization.

Step 3: Generalization (or Anonymization)

- Converted full date of birth into age (E.g. “2001-03-09” – “25.0”)
- Dropped the original birth date column

GDPR Art. 5(1)(c) - it keeps only the less detailed version of the data needed for analysis.

Step 4: Additional minimization

- Removed detailed spending behavior
- Noted that this field may have some business value,
- so its removal depends on necessity and justification.

GDPR Art. 5(1)(c) and **AI Act Art. 10**: unnecessary personal data should be minimized, and high-risk AI systems should use relevant, well-governed data. If the variable is not justified, removing it supports both.

04. Governance Recommendations

1

Audit Trails

- Record every credit decision with timestamp, model version, inputs used, outcome, and whether a human changed or confirmed the outcome.
- **GDPR Art. 5(2) (accountability)**
- **AI Act Arts. 11–12 (documentation)**

2

Human Oversight

- Model should support, not replace, human decision-making.
- Reduces risk of unfair automated decisions.
- **AI Act Art. 14**

3

Lawful Basis

- Document why each category of personal data is collected and used.
- Inform applicants about data use, purpose, and retention.
- **GDPR Arts. 6, 13**

4

Data Retention Policy

- Set retention periods for each data type.
- Delete or anonymize identifiers when no longer needed.
- **GDPR Arts. 5(1)(e), 17**

5

Access Control

- Restrict direct identifiers to authorized staff only.
- Analysts and data scientists should work mainly with pseudonymized datasets, not raw personal data.
- **GDPR Arts. 5(1)(f), 32**

6

Data and Model Review

- Review of each variable before use in analysis or modeling.
- Check necessity, justification, and proxy-risk.
- **GDPR Art. 5(1)(c); AI Act Arts. 9–10**

Key Findings

Data Quality

6 issue categories identified and remediated —> dataset reduced to 499 clean, validated records

Bias Detected

Disparate Impact Ratio of 0.76 (< 0.80) signals potential gender disparate impact requiring immediate model review

Privacy Gaps

6 Personal Identifiable Information fields found unprotected; Social Security Numbers and IPs pose critical GDPR and AI Act compliance risk

Governance

6 actionable controls proposed, from validation pipelines to fairness monitoring and human oversight